

Whittemore's Science and Practice *of* Pig Production

THIRD EDITION

Ilias Kyriazakis
BVet Med, MSc, PhD, MRCVS

Colin T. Whittemore
NDA, BSc, PhD, DSc, FIBiol, FRSE



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Preface and Acknowledgements

The first edition of *The Science and Practice of Pig Production* was by far the most extensive book that I had written. It was the result of a challenge that I should prepare a book of adequate scale and breadth to deal comprehensively with the whole international spectrum of the science and practice of pig production. The temptation to gather a consortium of authors and edit a multi-author title was resisted. I wanted this book to have cohesion and purpose, and to demonstrate an integrated philosophy. I wished to take the reader through the subject, examining the knowledge offered from a perspective that, although multi-dimensional, nevertheless sprang from a coherent and individual source. This could only be done by single authorship.

The second edition of the book had substantive additions, a good number of alterations and a few deletions. The opportunity to update had been timely, as much has happened in pig production and science since the first edition was written, but perhaps especially in biotechnology, meat and carcass quality, the understanding of growth, animal welfare, genetic improvement, disease and health, and some aspects of feeding and nutrition. Interestingly, the same subjects tend to figure in the major updates of this third edition.

For the third edition I decided that I could no longer support the contention that a single author could cover the scope of this book, and so a panel of international authorities was invited to join me in revising the text. In order to maintain the cohesion of a single text, and keep faith with the original principles of the book, I decided that the book should have a senior editor (Ilias Kyriazakis). The recognition given to all those who have contributed to the third edition is detailed on the Contributors page. I would like to acknowledge here the support and efforts of Professor Kyriazakis and the new contributors.

Although it remains the case that much of the work in the body of this book was completed over more than 30 years of research at Edinburgh, the new text now contains substantial bodies of information from other international centres, and this is recognised by the new panel of authors. One author cannot possibly have complete knowledge in breadth and in depth. For all of us, this dilemma can only be solved through learning from others and benefiting from their counsel, and this we have all done over many years of production and industry practice, university learning, scientific research and consultancy. We are pleased to be able to acknowledge formally the great debt we owe to all those from whom we have accumulated information, knowledge and wisdom.

The precious resource of input from the named authors, and all those around them, has nonetheless been subject to interpretation and to filtration. Such is the consequence of editing a large book. The editors, therefore, but especially myself, must bear responsibility for the final content.

Colin Whitemore
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Chapter 1

Introduction

The prime agricultural purpose of the pig is to provide human food (fresh, cured or processed), so pig production is necessarily preoccupied with the quality of the pig meat and the efficiency of its production. The pig grows rapidly, reproduces at a high rate and is accommodating in its eating habits. Historically, the pig was often seen as a scavenger; a rural scavenger in fields and woodlands, and an urban one using human food by-products and waste. These roles have ensured that the pig has been used as a domesticated agricultural animal widely throughout the world, but usually in small population sizes such as would facilitate by-product utilisation and encourage simple husbandry practices.

Recent history has seen the demand for high-quality pig meat products rise world-wide due to the increasing world population and improving human nutritional aspirations. This has raised the profile of the growth and reproductive characteristics of the pig. Simultaneously, improvements in other branches of agricultural practice have made available substantial quantities of cereal, legume and oilseed grains not appropriate – or not needed – for human consumption. The industrialisation of human food preparation has added to the changing environment by moving the availability of food by-products and waste away from the local and diversified domestic level and toward centralised manufacturing points. There has been a change in the nature and objectives of pig production over recent decades. Especially there has been created a large-scale integrated agricultural industry, producing substantial quantities of high-quality meat products which may be readily traded within and across international boundaries. The agricultural and biological sciences have been well set to serve pig production over its recent evolution, and the result has been a staggering example of the successful application of science to industry to the benefit of humankind. The modern pig industry bears little or no resemblance to that of only a few decades ago; either in its structure or in its method of trading, or in its definitions of quality in the end-product, or in its breeding, feeding, housing, health and management.

The major changes are often perceived in terms of sophistication and scale; and associated with the latter, intensification. Coping with these changes has demanded of science – and science has provided – new husbandry practices, new understandings of growth, reproduction and health, new appreciations of welfare and the environment, new nutritional approaches, and modern genetic improvement techniques. However, above all has been the essential placing of the application of science in

the context of latter-day production and marketing. Biological response is interpreted in terms of not only physical outcomes but also financial consequences.

Numerous small herds (<100) of breeding sows abounded as recently as two decades ago, but the major activities of pig meat production have now come to lie with the big integrators, often backed by corporate money, and where breeding, production, marketing, processing and consumer sales are closely coordinated. It is difficult to achieve efficiency of modern intensive pig production with unit sizes of less than 250 breeding sows, while 500+ would be unexceptional. Some of the larger operations will have a number of standard-built units spread about a locality, and managed from a central company headquarters (as, for example, in the US Carolinas). North America, originally the home of the family-sized hogg farm, now leads the big operators, with many companies having more than 10000 breeding sows and a significant number more than 100000. This pattern is followed now in Spain (where two companies each have around 100000 sows and each grow for slaughter 2 million or so meat pigs annually), Italy and Asia; while in the other European countries many pig producers remain individual owner-managers, albeit with units of 200–5000 sows.

While trends in pig production have had some degree of commonality worldwide, there remains diversity in international production practice and end-product objectives. Thus, although a high proportion of the sciences cognate to pig production are transferable across national barriers, the *way* the scientific information is used may need to vary widely. The most useful aspects of the scientific information base are those that address basic causes of biological and economic response, and *not* those that lay out the responses themselves in the form of rules and recommendations.

Throughout, the book argues causality and logic before presenting outcomes, and tries to avoid the bald presentation of the results of this or that production investigation as if the latter were to embody some form of universal truth; which of course it would not. It is fundamental to an international text that along with the description of phenomena must go their understanding. Universality lies in the depth of the understanding of biological and economic processes, not necessarily in the accuracy of their description. Nevertheless, pig production is a quantitative as well as a qualitative activity. Meat of given quantity and definable characteristic requires to be produced from feedstuffs of quantifiable nutritional value when given to specific pig genotypes in describable environmental, managerial and economic circumstances. Where possible, therefore, factors impinging upon the production process are expressed in hard and quantifiable terms. After all, it is only through quantification that the absolute and relative importances of the various aspects of the sciences and practice of pig production can be compared, judged and ultimately used.

The world's pig industry is a success story founded in the application of science to management. The extent of the success thus far is well illustrated in Figures 1.1 and 1.2. However, there is no reason for complacency, and a great deal more can be done to improve the level of efficiency of output and the quality of the end-product.

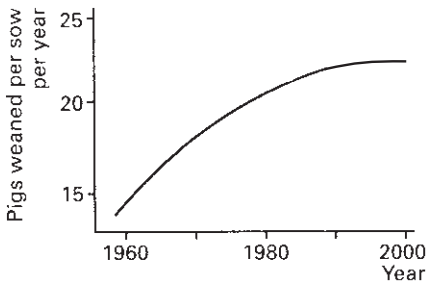


Fig. 1.1 Rate of increase in sow productivity (northern European data).

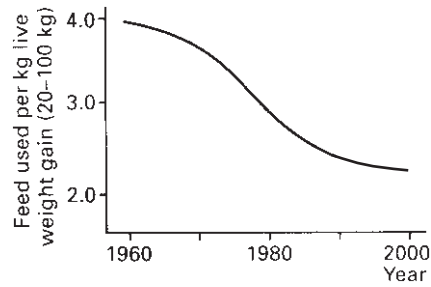


Fig. 1.2 Rate of improvement in feed usage per kilogram live weight gain. The 1970s saw the introduction of high nutrient density diets in significant quantities (northern European data).

Most pig production enterprises world-wide are under-achieving, and there is a considerable disparity in production efficiency between different nations. Compared to the presently known potential, the world's pig industry falls far short. Application of scientific principles and sound production practices can redress that short-fall.

Chapter 2

Pig Meat and Carcass Quality

Introduction

World pigmeat production has risen linearly in the latter half of the twentieth century, with some authorities estimating a doubling in output over this time. At present, 50% of pig production is from Asia and the Pacific (one-third of the world's pigs are in China), 30% from Europe and the former USSR, 13% from North America and the remaining 5% from Africa and South America. The world's expansion has taken place almost entirely in Asia and the Pacific (over 300%), expansions in Europe, the former USSR and North America being only around 40% since 1970. The popularity of pig meat differs widely throughout the world. Of all the meat eaten in Scandinavia 60% is pig meat, in the European Community 50%, in Japan 45% and in North America 35%, but in Argentina only 5% of all meat eaten is pig meat. World-wide, pig meat is the most popular of all available meats.

It may be taken for granted that pig meat product must first be *needed* (i.e. there must be a market), and next be safe to eat. It must also be *efficiently produced* (competitively priced) and at the required level of *quality*. However, increasingly meat production must satisfy further criteria in relation to production *method* and production *environment*, which must be both *sustainable* and *ethical*.

Within recent history, the pig has been bred for both supreme fatness (the required commodity being lard) and supreme leanness (the required commodity being muscle). Whether the acceptable pig carcass be fat or lean depends much upon national predilection. As industrialisation develops, the desire for lean meat appears to dominate the definition of carcass quality. Eastern European, Japanese and North and South American tastes are, like Western Europe, demanding progressively lower levels of meat fat content. Asian pigs, numerous in China and the Pacific countries, are overfat by Western standards, and it is likely that some desire to reduce fatness will take place there as well. Nevertheless, one nation's lean pig remains another nation's fat pig.

In Europe, the pig meat market used to be for mass product at low price. Indeed, general advancement of knowledge in pig production has resulted in the real cost of pig meat now being a mere 30% of what it was 40 years ago. There are now 12000000 sows in the European Union and 120000000 pigs in total. The average European consumption of pig meat per head per year is more than 40 kg. There is substantial cross-border trade in pig products, a trend that will increase if the present export ambitions of some countries (including the USA) come to be realised. In the

UK there are 800 000 sows and 8 000 000 pigs in total. Fifteen million pigs are slaughtered annually, producing some 1 000 000 tonnes of pork, bacon and ham product. Half of the bacon eaten in UK is imported, and about half of that comes from Denmark, with a quarter from the Netherlands and a further quarter from the Irish Republic. The UK exports about the same amount of pork as it imports (100 000 tonnes), but the importation of bacon and ham is some 350 000 tonnes, and there is no effective export trade. The UK public consumes 15 kg per head per year of fresh pork and 7 kg per year of bacon and ham. There is also further consumption of pork through prepared pig meat products. Italy imports substantial quantities of meat products from pigs slaughtered at live weights of below 120 kg but exports substantial quantities of meat products from pigs slaughtered at live weights above 150 kg. Japan has been traditionally supplied from Taiwan, but Denmark and the USA have recently expanded their export business to Japan.

Fat reduction, now readily achievable to its optimum level, was initiated originally not by consumer discrimination against fat but by the economics of pig production and pig processing, both of which benefit from a lean carcass. Pig markets still tend to divide into two types: one uses pig meat as a substrate for a huge variety of manufactured meat products and satisfies a continuing need for low-cost meat; the other, of increasing importance, attends to the consumer supply of pork and bacon meat of high eating quality. This latter market needs carefully prepared cuts, a low level of subcutaneous fat, an adequate level of intramuscular fat and a large mass of lean tissue, giving a high perception of eating pleasure – meat that is juicy, tender and tasty, avoiding all suggestion of dryness, paleness, taint or toughness. It should not be presumed, however, that a demand for lean pig meat is universal. Indeed, the production of lean genotypes for the northern European pig meat consumer can be the cause of a need for remedial action in southern Europe and in Asia, where a much higher level of fat is required – at least for the present.

Producers may be reluctant to accept the restraints imposed on them by a meat trade concerned primarily with product quality. Conversely, the trade may be slow to allow producers to profit from the benefits of new techniques and streamlined production practices. Such problems will remain with an industry which in many countries continues to be based upon adversarial trading and mistrust at each interface of the production chain. The successful future of pig industries the world over is likely to require the benefits of integration between breeder, producer, feed compounder, meat packer and retailer. The producer needs effective lines of communication with the market. If that communication generates information that is useful, both producer and meat trader will profit.

Marketing

Europe is an example of an industrialised economy with primary agriculture contributing 2–5% of Gross Domestic Product (GDP) and employing 3–8% of the work force. The livestock sector comprises some 50–60% of total farm income. Food and

drink manufacture represents a surprisingly large 10–15% of the manufacturing sector, contributes at least 2% of GDP and employs around 3% of the work force. Twelve percent of overall consumer expenditure is on food, being some 20% of household expenditure. Of the total food expenditure, some 40% comprises meat, dairy products and eggs, of which some one-fifth of the cost can be traced back to the primary product, four-fifths being added value in the manufacturing, processing and retail sectors. Half of the meat purchased by consumers is derived from pigs, and the trend continues to favour pig meat consumption over other meats.

Live pigs are marketed at any time subsequent to weaning (which can be between 6 and 20 kg and at 3–8 weeks of age). Pig meat can come from suckling pig at one extreme, and from 180 kg animals destined for the production of large Italian Parma hams at the other. National consumption patterns cause pigs to be finished and consumed at a wide variety of weights between these extremes. As the mature size of many Asian pig types is so much less than European types, slaughter weights are usually less. In many European countries 50–80 kg live weight is the traditional *butcher* range for the production of small, fresh pork joints bought and consumed locally, while in Japan, Australia, North and South America and all the countries of Europe 90–120 kg live weight is the traditional range for *industrial processing* at meat packing plants which distribute fresh pork joints, cured bacon products and delicatessen lines. Animals used for the production of hams which are dry-cured by slow and natural processes (such as prosciutto in Italy and Serrano in Spain) are usually finished to substantially greater live weights. Cull sows weighing 120–300 kg may command a useful price according to the state of the market. Meat from cull sows can be satisfactorily incorporated into a range of products, and it can be cost-effective to sell culled animals in good body condition when the price is satisfactory. Adult male pigs that have been used for breeding are inappropriate for human consumption, the meat being coarse and often tainted. This criticism does not apply to fast-grown entire males of European breed types slaughtered at less than 110 kg.

Prime meat is usually found at around 30–60% of mature size (Figure 2.1). At less than 30% there is a tendency toward lack of flavour, whilst at greater than 60% there is a tendency toward lack of tenderness. Pig types of larger mature size merit a higher slaughter weight, and it is noticeable that conventional slaughter weights in the European pig industry have been increasing over recent years. Most pigs used for processing and meat packing in industrialised countries have a mature weight

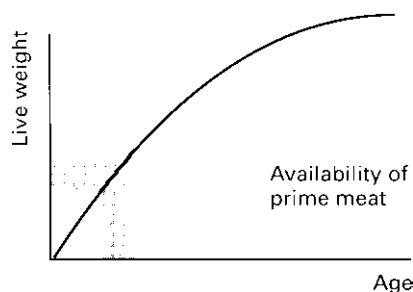


Fig. 2.1 Prime meat is found from pigs slaughtered at between 30 and 60% of mature size.

of between 250 and 350 kg, while in Asian types mature weight can be as low as 150 kg.

The primary pig production industry may segregate into specialised farm types; often breeders or feeders. Specialised breeding units will move weaned or grower pigs at weights of 7–10 kg (weaners) or 20–40 kg (growers) to grow-out units, which will feed the weaners to the next stage or, in the case of growers, to finished weight. Such systems allow concentration of resources, skills and preferences. Frequently, breeder units are small, being part of mixed family farming enterprises. Grow-out units, on the other hand, tend to be larger and more industrialised, and may buy from many sources of supply. Such specialisation may lead to more problems than solutions, unless the ill-effects of transport and trading in young animals are avoided, variability in type of pig supplied is minimised and disease incursion eschewed. Modern enterprises are best structured either (1) as breeder/finisher operations where a single unit both breeds and grows the pigs to final specification or (2) as a multi-site operation under single management control integrated at each step. Single-step operations also allow a more effective relationship between ultimate customer and primary producer; thus meat quality aspects as perceived by the meat packer and retailer may be effectively relayed back to the breeder. Choice of pig type or hybrid by the breeder is of vital interest to grower, packer and retailer. Any discontinuity of production flow, such as an industry based on separate businesses dealing with the breeding and feeding aspects of the production process, can only serve to obscure an optimum policy and hold back the development of the industry.

Within any one nation, or group of inter-trading nations, the pig industry has potential for expansion or contraction. The extent of this potential is directly linked to the number of producers who have alternative sources of income available to them. When pig prices are good the size of the breeding herd is expanded and new herds are begun. This may be achieved readily by retaining for breeding purposes female stock originally destined for slaughter, or by the purchase of extra hybrid breeding stock replacements. These animals will produce young some 6 months later, and a further 6 months downstream additional slaughter pigs become available for sale. At this point supply is enhanced and prices will ease back. The elasticity of demand for pig products is such that although customers will purchase more pig meat as prices fall and it becomes cheaper, the ability to eat more does not keep pace with the falling price. Pig prices will continue to fall for approximately a further year or so, at which point breeding herds will begin to be reduced in both size and number. This process may take another year before a shortage of available slaughter pigs becomes evident and prices begin to strengthen again. This simplistic pig cycle (Figure 2.2) is obfuscated by fresh meat butchers equalising prices of various meats, and also by fluctuations in the world supply of pig feed ingredients. Grain and soya prices are relatively independent of pig price because grains are used for purposes other than the feeding of pigs. When pig feed ingredient prices are low, producers can tolerate a reduced pig carcass price for a longer time before needing to retract. A combination of favourable international grain prices and

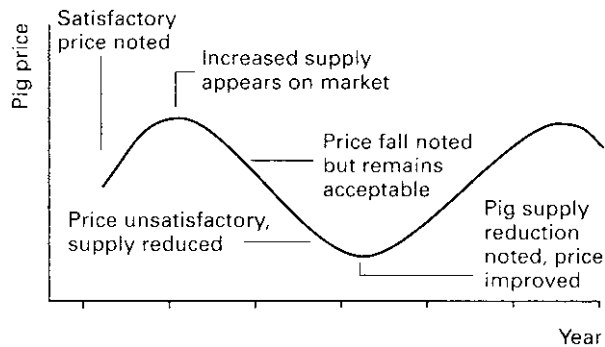


Fig. 2.2 The cyclical nature of pig prices. These are further subject to influence from the price of competitive meats, prevailing feed price and the availability of capital.

favourable pig carcass prices will produce a profit bonanza and an accelerated rate of expansion.

The ebb and flow of the pig cycle as a result of pig and grain prices is more apparent in countries where capital expenditures do not figure largely. For an industrial-based intensive pig industry with high capital investment in its buildings, the availability of money at satisfactory interest rates acts as a regulator on the entrepreneur's ability to either expand when times are good or contract when they are not.

Consumption patterns

The estimated world consumption of animal products (millions of tonnes) is as follows: milk 400 (steady), pig 63 (rising), beef and veal 48 (steady), poultry 33 (rising), eggs 20 (steady), farmed fish 8 (rising), sheep and goat 6 (steady). Pig meat accounts for more than 40% of the world's meat consumption and is produced from about 850 million pigs (about 60 million breeding sows). The European Community has about 120 million pigs from which come 14 million tonnes of pig meat. This gives the European Community more than 20% of world pig meat production. Within the Community more than 30% of pig meat production comes from Germany, about 12.5% from each of France, the Netherlands, Italy and Spain, while the UK and Denmark produce around 10% each. About 25% of pig meat consumed in the European Community is eaten fresh, the rest being cured or processed into products with a greater or lesser degree of added value.

Notwithstanding some fat being necessary in meat to allow optimum meat quality (particularly with regard to succulence, taste, texture and cookability), the report of the UK Committee on Medical Aspects of Food Policy relating to diet and cardiovascular disease makes specific mention of the need for a reduction in the level of fats in the human diet. The incidence of heart disease is commonly thought to be reduced by increasing the ratio of polyunsaturated fatty acids to saturated

fatty acids in the diet. It would appear that saturated fatty acids with less than 18 carbon atoms in the chain are the most likely to increase the risk of coronary heart disease. Some 40% of the total energy in the diets of many people in the industrialised world comes from fat of vegetable or animal origin. The proportion of total dietary energy from fats has been rising inexorably over recent years; and it is perhaps this simple statistic that is the more relevant. Meanwhile, there has been a simultaneous dramatic reduction of fat in pig meat, down from about 30% of the total carcass in the 1960s to about 15% or less in an 80 kg carcass of today. Subcutaneous backfat measurements have been reduced by genetics, nutrition and feeding management by about 0.5 mm annually. Pig subcutaneous fatty tissue contains about 70% lipid, and two thirds or more of all fat on a pig carcass may be found in the subcutaneous depots. For top grade bacon and pig meat products, backfat depth targets at the P2 site for pigs of 90 kg live weight can be as low as 10 mm. Given the potential of pig meat to be extremely lean, it may be assumed that on health grounds alone the consumption of pig meat and pig products will continue to further increase.

Medical wisdom advises that fat consumption should fall to around 30% of total diet energy, and that the polyunsaturated fatty acid:saturated fatty acid ratio in dietary fat should be around 1.0. Presently in many countries of the Western world the ratio is about 0.3. The Committee on Medical Aspects of Food Policy has suggested 0.45 as an achievable target. As about half of all diet fat comes from animal products (including milk, milk products, eggs and meat), and as some meat fats have a poor polyunsaturated:saturated fatty acid ratio, there may be consumer benefit in the identification of meat and meat products low in fat, but what fat there is being high in polyunsaturated fatty acids, as is the case for pig meat. The ratio of polyunsaturated:saturated fatty acids in pig meat is a good 0.7.

The polyunsaturated fatty acids in the meat are of two distinct types: the *n*-6 (omega-6) fatty acids including linoleic acid (18:2) which reaches the highest concentration of all the polyunsaturated fatty acids and the *n*-3 (omega-3) fatty acids of which α -linolenic acid (18:3) has the highest concentration. (The nomenclature *n*-6 or *n*-3 refers to the location of the double bonds in the chain. In '18:2', etc. the number before the colon refers to the number of carbon atoms in the chain and the number after the colon refers to the number of double bonds. The more double bonds, the more unsaturated the fat.) 18:2 and 18:3 are precursors of the 'long chain polyunsaturated fatty acids' of the *n*-6 and *n*-3 series respectively. These are metabolically important compounds, being constituents of cell membranes and substrates for the formation of eicosanoids which are regulatory molecules. A ratio of *n*-6:*n*-3 polyunsaturated fatty acids in the diet of less than 4.0 is recommended and most Western diets are much higher than this. The value in pig meat is around 7.0, too high. This is due to a high level of 18:2 which is the major fatty acid in the grains and oilseeds used for producing pig diets. Pigs are very efficient at transferring dietary fatty acids into muscle and fat tissue. The high polyunsaturated:saturated fatty acid ratio in pig meat is due to this efficient transfer of 18:2, but on the other hand this upsets the *n*-6:*n*-3 ratio. Research at Bristol University has shown that the

balance can be corrected using whole linseed or linseed oil. This plant is unusual in having a high 18:3 content. Likewise, the valuable long-chain *n*-3 polyunsaturated fatty acids can be increased in meat by feeding fish oils to the pig. This can produce higher levels than those formed naturally from 18:3 in a series of enzymatic steps.

Muscle from lean pigs is very low in fat, around 3–5% in muscle defatted on the plate and around 1% in muscles from which all carcass fat is removed. This is the true intramuscular lipid or marbling fat. Muscles differ in marbling fat levels, with the loin muscle (*longissimus dorsi*) being relatively high in fat compared with other muscles in the carcass. The fat tissue in lean pigs is also high in 18:2. This is because dietary fat forms a high proportion of body fat in lean pigs, compared with the pig's own synthesis of saturated and monounsaturated (one double bond) fatty acids. In conclusion, modern pig meat is low in fat and high in polyunsaturated fat and therefore makes an important contribution to a healthy diet. We also have the means to raise *n*-3 fatty acids in pig meat.

People in most advanced European countries eat about 70–90kg of total meat per person per year; this is about 30% less than in the USA. In the UK much less pig meat is eaten, in both absolute and proportional terms, than in Denmark, Germany or the Netherlands (Table 2.1). In the UK the consumption of imported cured pig meat and pig meat products accounts for 30–40% of total consumption and comes from a variety of sources, including Denmark, the Netherlands, Poland and Germany.

The annual average meat consumption per person per year in the European Community is 5kg of sheep meat (falling), 25kg of beef meat (steady), 15kg of poultry (rising) and 40kg of pork and bacon (rising). The European Community is 100% self-sufficient for pig meat overall. The total tonnage of production from the European Community is 14 million tonnes, the major contributors being Germany, the Netherlands, France and Spain. The European Community has around 12 million sows from which this production comes.

In terms of value for money, pig meat is a better buy than beef or lamb, and rivals chicken and turkey. Pig meat is highly versatile, being used for fresh meat joints, cured joints, bacon, cooked meats (fresh and cured), processed products, pâté, barbecue ribs and many specialised lines of sausages, delicatessen items and so on.

The efficiency of transfer of animal feed into pig meat is high (Figure 2.3) and continues to improve with new production technology. Increasing the range of pork and bacon products available to consumers (Figure 2.4) is probably the most appropriate way now open for further expansion of the pig meat sector.

Quality assurance

In the middle years of the twentieth century, food shortages in many developed countries, the need to save on imports, and successive governments' policies to research and produce an ample supply of subsidised food led to dramatic increases in the output of agricultural product (including pigs) from farms. The social and eco-